FE5209 | Homework Assignment 1

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1. Abbott Laboratories (ABT) and Market Index Return Analysis

1.1 Descriptive Statistics of Simple Returns

- **Objective:** Compute the sample mean, standard deviation, skewness, excess kurtosis, minimum, and maximum for ABT stock, CRSP value-weighted index (VW), CRSP equal-weighted index (EW), and S&P composite index.
  R Commands Used:
  `basicStats()`
- **Note:** BasicStats() returned Excess Kurtosis with row name Kurtosis. Changed the name of the row to Excess Kurtosis for the returned value.

  Results:

	abt	vwretd	ewretd	sprtrn
Mean	0.014073	0.009020	0.011583	0.006386
Stdev	0.064795	0.046341	0.057299	0.044788
Skewness	0.097058	-0.555341	-0.200025	-0.444303
Excess Kurtosis	2.469971	2.073657	3.288104	1.889777
Minimum	-0.234146	-0.225363	-0.272248	-0.217630
Maximum	0.382326	0.165585	0.299260	0.163047

1.2 Descriptive Statistics of Log Returns

- **Objective:** Transform simple returns into log returns and compute the same statistics.
  R Commands Used:
  `log()` & `lapply()` for transformation, `basicStats()`
- **Note:** BasicStats() returned Excess Kurtosis with row name Kurtosis. Changed the name of the row to Excess Kurtosis for the returned value.
  Results:

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	abt	vwretd	ewretd	sprtrn
Mean	0.011924	0.007903	0.009887	0.005360
Stdev	0.064287	0.046745	0.057491	0.045145
Skewness	-0.297125	-0.839542	-0.662276	-0.711215
Excess Kurtosis	2.005416	3.003331	3.976081	2.659018
Minimum	-0.266764	-0.255361	-0.317795	-0.245428
Maximum	0.323768	0.153223	0.261795	0.151043

1.3 Hypothesis Testing on ABT Log Returns

- **Objective:** Test the null hypothesis that the mean of the log returns of ABT stock is zero.
  R Commands Used:
  `t.test()`

  Testing:
  Null Hypothesis (H0): The mean log return is zero.
  Alternative Hypothesis (Ha): The mean log return is not zero

  Results:
  1. We **reject the Null Hypothesis** because the *p-value (0.00004555) is significantly lower* than the *alpha level of 0.005* for a two-tailed test at the 1% significance level
  2. The sample mean of the log returns is 0.0119, suggesting that the *average log return of ABT stock is* **positive and statistically different from 0**

```
        One Sample t-test

data:  log_returns$abt
t = 4.1143, df = 491, p-value = 4.555e-05
alternative hypothesis: true mean is not equal to 0
99 percent confidence interval:
 0.004429679 0.019418891
sample estimates:
 mean of x
0.01192429
```

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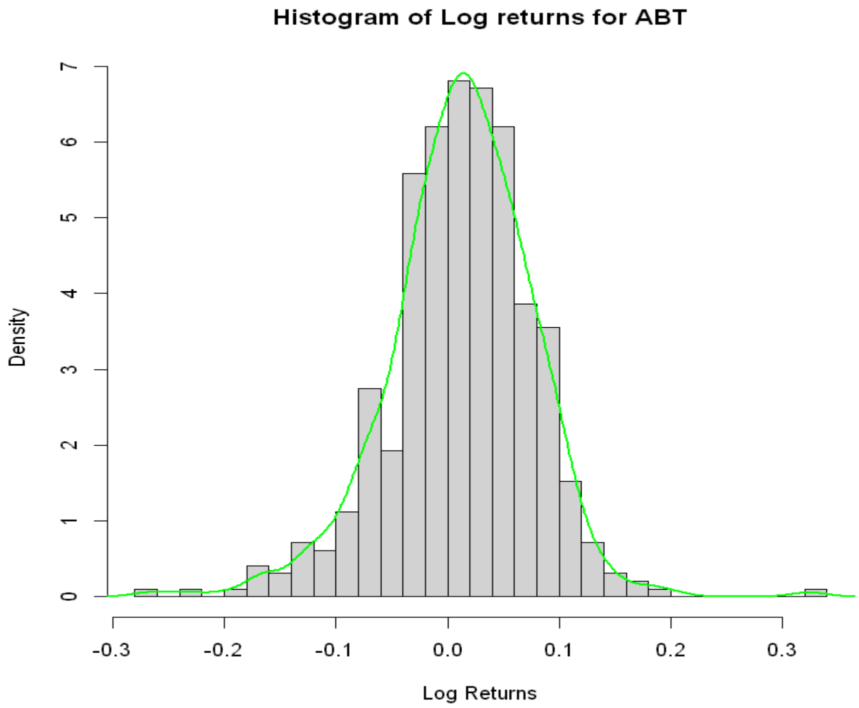
1.4 Histogram and Sample Density Plot of ABT Log Returns

- **Objective:** Visualise the distribution of ABT's log returns.
  R Commands Used:
  `hist()` for histogram, `density()` for sample density plot

  Results:



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2. CRSP Value-Weighted Index (VW) Return Analysis

2.1 Hypothesis Tests on Monthly Returns

Objective: Perform various hypothesis tests on the monthly returns of Value Weighted Index (VWI)

(a)

R Command Used:
`t.test()`

Testing:
Null Hypothesis (H0): The monthly mean returns of VWI is 0
Alternative Hypothesis (Ha): The monthly mean returns of VWI is not zero

Results:

1. We **reject the null hypothesis** because the *p-value (0.00001912) is significantly lower than the alpha level* of 0.005 for a two-tailed test at the 1% significance level
2. The sample mean of the *VW simple returns is 0.0090*, suggesting that the average return of the value-weighted index is **positive and statistically different from zero**

```
	One Sample t-test

data:  vw_simple_returns
t = 4.3172, df = 491, p-value = 1.912e-05
alternative hypothesis: true mean is not equal to 0
99 percent confidence interval:
 0.003617125 0.014421948
sample estimates:
  mean of x
0.009019537
```

(b)

R Command Used:
`pnorm() & qnorm()`

Testing:
Null Hypothesis (H0): The skewness (m3) of VWI data is 0
Alternative Hypothesis (Ha): The skewness (m3) of VWI data is lesser than zero

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Results:

1. We **reject the null hypothesis** because the p-*value (2.277105e-07) is significantly lower than the alpha level* of 0.01 for a one-tailed test at the 1% significance level

2. The skewness of the *value-weighted index monthly stock returns is -0.5570*, indicating that the **distribution is negatively skewed**

```
Skewness: -0.5570388
H0 : m3 = 0 versus Ha : m3 < 0, where m3 denotes the skewness.
We reject the null Hypothesis (H0).
Test statistic (S*): -5.044201
p-value: 2.277105e-07
```

(c)

R Command Used:
pnorm() & qnorm()

Testing:
Null Hypothesis (H0): The kurtosis (K) of VWI data is 0
Alternative Hypothesis (Ha): The kurtosis (K) of VWI data is greater than zero

Results:

1. **We reject the null hypothesis** because the *p-value (0) is significantly lower than the alpha level* of 0.01 for a one-tailed test at the 1% significance level

2. The kurtosis of the *value-weighted index monthly stock* returns is 5.0943, which is higher than 3, indicating that the **distribution has heavier tails and is more peaked compared** to a normal distribution

```
Kurtosis: 5.094344
H0 : K = 3 versus Ha : K > 3, where K denotes the kurtosis.
We Reject the Null Hypothesis (H0)
Test statistic (K*): 9.482548
p-value: 0
```

(d)

R Command Used:
`t.test()`

Testing:
Null Hypothesis (H0): The monthly mean returns of VWI is 0
Alternative Hypothesis (Ha): The monthly mean returns of VWI is greater than zero

Results:

1. We **reject the null hypothesis** because the *p-value (0.000009561) is significantly lower than the alpha level* of 0.01 for a one-tailed test at the 1% significance level

2. The *sample mean of the VW simple returns is 0.0090*, suggesting that the average return of the value-weighted index is **positive and statistically greater than zero**.

3. The 99% confidence interval for the mean return is (0.0041, ∞), indicating that the **mean return is significantly greater than zero** at the 1% significance level

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```
        One Sample t-test

data:  vw_simple_returns
t = 4.3172, df = 491, p-value = 9.561e-06
alternative hypothesis: true mean is greater than 0
99 percent confidence interval:
 0.004143421         Inf
sample estimates:
  mean of x
0.009019537
```

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3. U.S. Real GDP Growth Rates Analysis

3.1 Plot of GDP Growth Rates

- **Objective:** Visualise the growth rates and observe the Great Moderation.
  R Commands Used:
  `plot()`

  Results:
  1. Evidently it can be seen from the graph that after 1980s the *volatility in GDP has reduced* as defined in the **Great Moderation**

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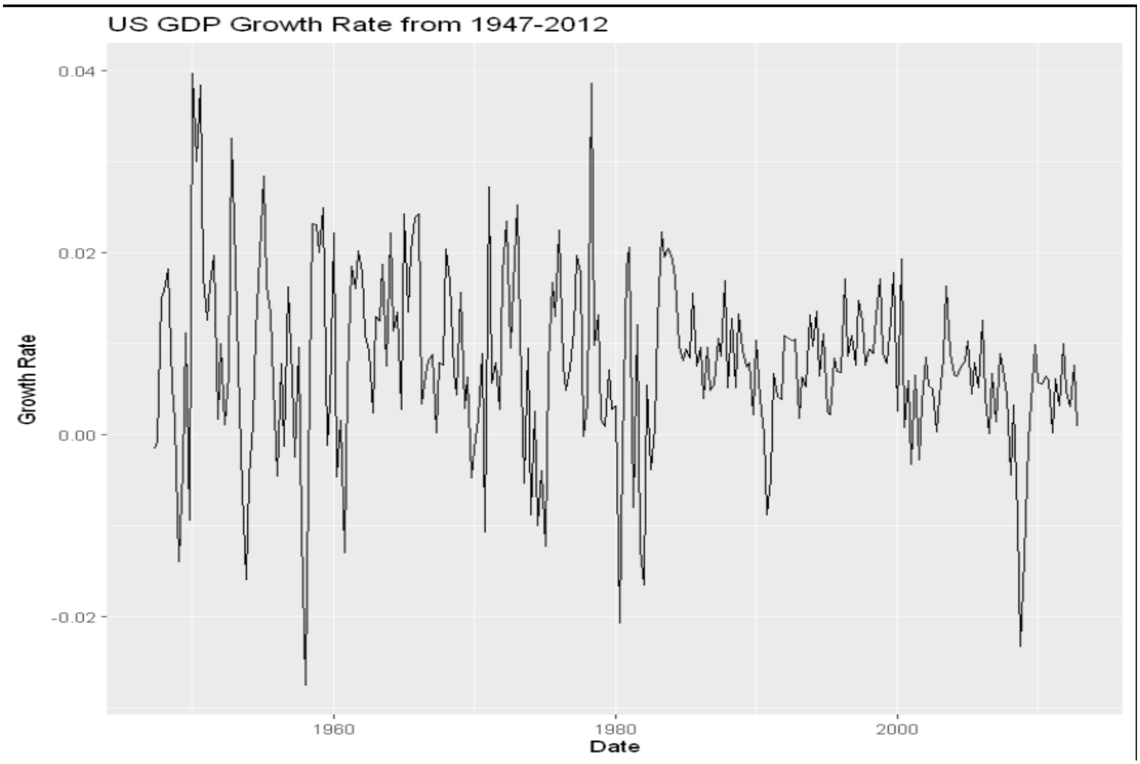

3.2 Autocorrelation Tests

- **Objective:** Test for autocorrelation in GDP growth rates
  R Commands Used:
  Box.test() *{Ljung-Box test for the first 12 lags}*

  Testing:
  Null Hypothesis (H0): ρ1 = ρ2 = ... = ρ12 = 0
  Alternative Hypothesis (Ha): ρi ≠ 0 for some i = 1,..,12

  Results:
  1. We **reject the null hypothesis** because the *p-value (0.000000003737) is significantly lower than the alpha level of 0.01*
  2. Thus, we conclude that the **lagged GDP growth rates are not independent** and that at least one of the lagged autocorrelation coefficients is significantly different from zero

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```
        Box-Ljung test

data:  data$growth_rate
X-squared = 64.259, df = 12, p-value = 3.737e-09
```

3.3 Testing the Mean of GDP Growth Rates

- **Objective:** Test if the mean of the U.S. GDP growth rates is zero.
  R Commands Used:
  `t.test()`

  Results:
  1. We **reject the null hypothesis** because the *p-value (< 2.2e-16) is significantly lower than the alpha level of 0.005* for a two-tailed test at the 1% significance level
  2. The *sample mean of the U.S. GDP growth rates is 0.0078*, indicating that the average GDP growth rate is **positive and statistically different from zero**

```
        One Sample t-test

data:  data$growth_rate
t = 12.786, df = 262, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
99 percent confidence interval:
 0.006193171 0.009346756
sample estimates:
  mean of x
0.007769964
```

3.4 Order of the AR Model

- **Objective:** Determine the order of the AR model for the growth rate series.
  R Commands Used:
  `ar()`

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Results:

1. The Order of the AR Model for the **Growth Rate Series is 3.**

```
# Fit an AR model to the growth rate series using MLE
ar_model <- ar(na.omit(data$growth_rate), method = "mle")

# Extract and print the order of the AR model
ar_order <- ar_model$order
cat(ar_order)

3
```

---

4. AR Model for GDP Growth Rates

4.1 Model Construction and Checking

- **Objective:** Build and validate the AR model for U.S. GDP growth rates.

  R Commands Used:

  `ar()`, `polyroot()` for checking

  Results:

  1. The fitted AR(3) model for the U.S. GDP growth rate series is given by:

$$Growth\ Rate_t = 0.0077 + 0.3476(Growth\ Rate_{t-1}) + 0.1282(Growth\ Rate_{t-2})$$

$$- 0.1122(Growth\ Rate_{t-3)} + \epsilon_t$$

Model Coefficients:

- **AR(1) Coefficient (ar1):** 0.3476 (s.e. = 0.0612)
- **AR(2) Coefficient (ar2):** 0.1282 (s.e. = 0.0644)
- **AR(3) Coefficient (ar3):** -0.1122 (s.e. = 0.0613)
- **Intercept:** 0.0077 (s.e. = 0.0009)

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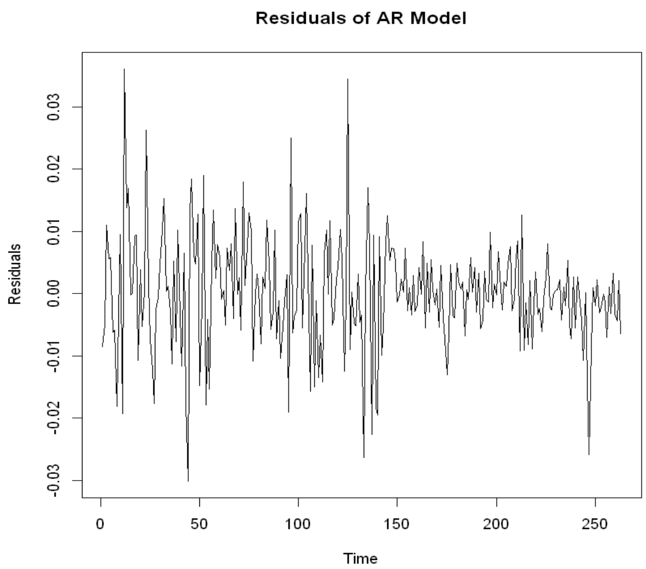
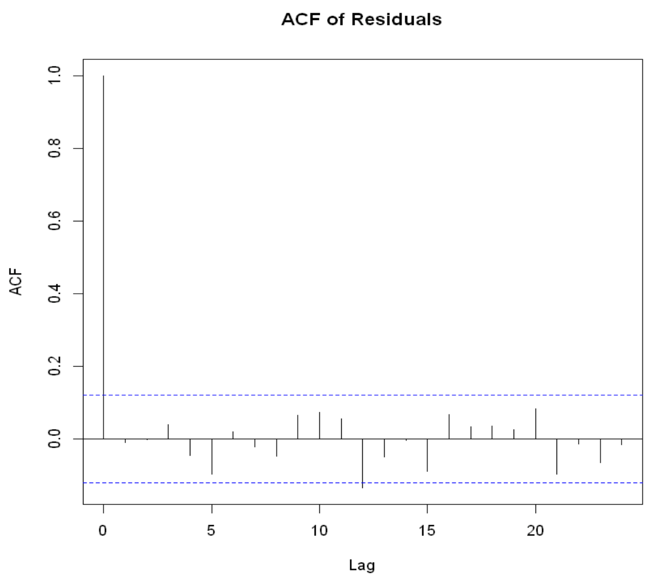
- **Estimated Variance:** 8.17e-05
- **Log Likelihood:** 864.45
- **AIC:** -1718.9
- ϵ_t is the white noise error term






```
arima(x = na.omit(data$growth_rate), order = c(3, 0, 0))

Coefficients:
         ar1     ar2      ar3  intercept
      0.3476  0.1282  -0.1122     0.0077
s.e.  0.0612  0.0644   0.0613     0.0009

sigma^2 estimated as 8.17e-05:  log likelihood = 864.45,  aic = -1718.9
```

Model Checking:

The Box-Ljung test was performed on the residuals of the fitted model:

- **Test Statistic:** X^2 = 19.99
- **Degrees of Freedom (df):** 20
- **p-value:** 0.4586

Results:

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1. Since the *p-value (0.4586) is greater than the alpha* level of 0.05, **we fail to reject the null hypothesis** that the residuals are independently distributed
2. This indicates that there is **no significant autocorrelation present in the residuals**, suggesting that the fitted *AR(3) model adequately captures* the dynamics of the GDP growth rate series
3. Hence, the **AR(3) model is a suitable representation of the growth rate series**, and the model validation checks confirm its reliability

```
        Box-Ljung test

data:  residuals(ar_model)
X-squared = 19.99, df = 20, p-value = 0.4586
```

4.2 Business Cycle Confirmation

- **Objective:** Check for the existence of business cycles.

  Results:
  1. The presence of complex roots (with a modulus greater than 1) in the AR(3) model indicates that the **model exhibits cyclical behaviour**. The roots suggest oscillations in the time series data, which corresponds to the **presence of business cycles**
  2. The calculated business cycle **frequency of 10.26219 quarters = approximately 3 years** indicates that the model captures cycles that are typical of economic fluctuations, supporting the conclusion that the model confirms the existence of business cycles in the U.S. GDP growth rates
  3. In summary, the characteristics of the roots show that the **AR(3) model adequately reflects cyclical**

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behaviour, thereby confirming the existence of business cycles in the data

```
[1]  1.657814+1.164277e+00i -2.172531+5.506659e-15i  1.657814-1.164277e+00i
Modulus of the roots: 2.025805 2.172531 2.025805
Business cycle frequency: 10.26219  quarters =  3 years (approx)
```

4.3 Forecasting GDP Growth Rates

- **Objective:** Generate 1-step to 8-step forecasts with 99% confidence intervals.
  R Commands Used:
  `predict()`

  Results:
  1. Overall, the forecasts suggest a **cautiously optimistic outlook for U.S. quarterly GDP growth**, with the potential for slight fluctuations in growth rates over the next eight quarters

	Step	PointForecast	LowerBound99	UpperBound99
1	1	0.005880335	-0.01740250	0.02916317
2	2	0.006228356	-0.01842120	0.03087791
3	3	0.007733294	-0.01758914	0.03305573
4	4	0.007747671	-0.01757863	0.03307397
5	5	0.007906583	-0.01741972	0.03323288
6	6	0.007794875	-0.01753847	0.03312822
7	7	0.007774803	-0.01755986	0.03310947
8	8	0.007735680	-0.01759953	0.03307089

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5. Comparison of AR(1) and AR Models

5.1 AR(1) Model Fitting

- **Objective:** Fit an AR(1) model to the GDP growth rate series.
  R Commands Used:
  `ar()`

  Results:
  1. The fitted AR(1) model for the U.S. GDP growth rate series is given by:

$$Growth\ Rate_t = 0.0077 + 0.3705(Growth\ Rate_{t-1}) + \epsilon_t$$

Model Coefficients:

- **AR(1) Coefficient (ar1):** 0.3705 (s.e. = 0.0572)
- **Intercept:** 0.0077 (s.e. = 0.0009)
- **Estimated Variance:** 8.344e-05
- **Log Likelihood:** 861.71
- **AIC:** -1717.43
- ϵ_t is the white noise error term

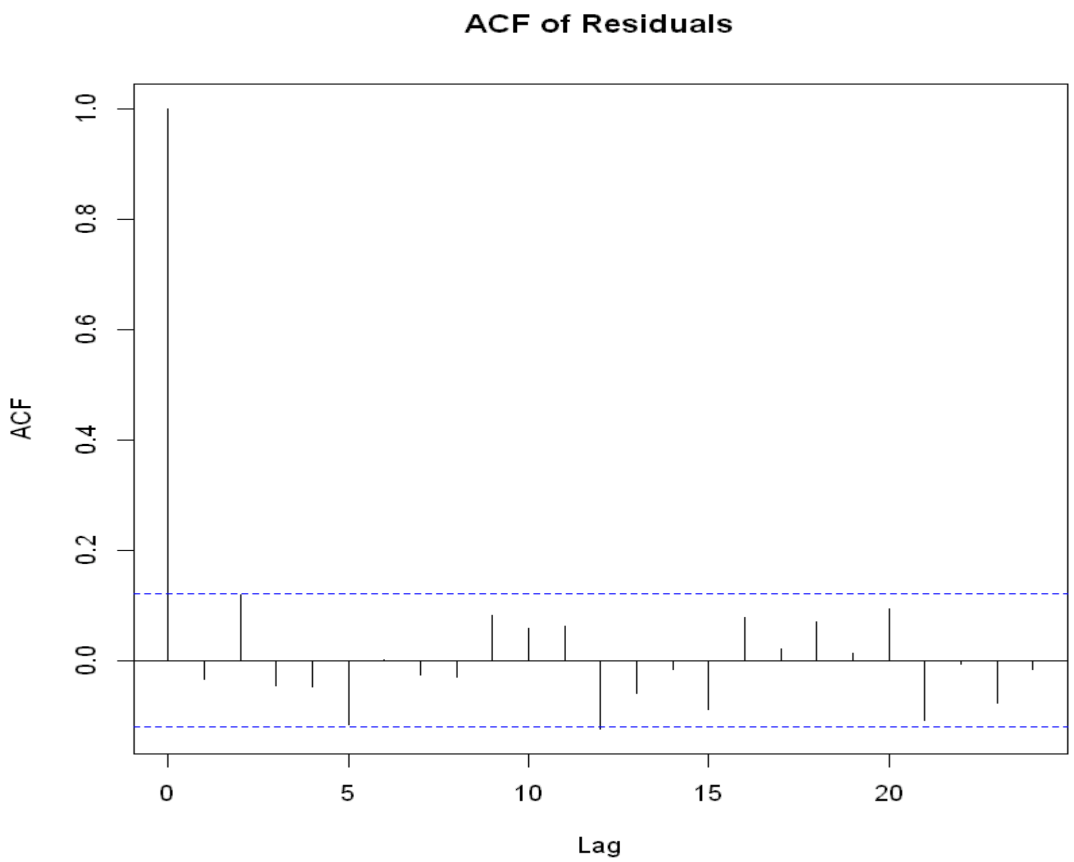
```
Call:
arima(x = na.omit(data$growth_rate), order = c(1, 0, 0))

Coefficients:
         ar1  intercept
      0.3705     0.0077
s.e.  0.0572     0.0009

sigma^2 estimated as 8.344e-05:  log likelihood = 861.71,  aic = -1717.43
```

5.2 AR(1) Model Adequacy

- **Objective:** Check if the model is adequate.
  R Commands Used:
  `acf() Box.test()`

  Results:

  1. The Ljung-Box test results in a p-value of 0.1487. Thus, we fail to reject the null hypothesis at a 1% significance level. Implying residuals are white noise, so the AR(1) model is adequate.




```
        Box-Ljung test

data:  residuals(ar_model)
X-squared = 26.539, df = 20, p-value = 0.1487
```

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5.3 AR(1) vs AR(3)

- **Objective:** Compare models for preferred.
  R Commands Used:
  `AIC() BIC()`

  Results:

  1. As both models are adequate. Compare the models based on the AIC and BIC values. The smaller the value the better. The corresponding values for AR(1) and AR(3) are
     a. AIC for AR(3): -1718.903
     b. BIC for AR(3): -1701.042
     c. AIC for AR(1): -1717.43
     d. BIC for AR(1): -1706.713
  2. Based on AIC AR(3) is preferred and AR(1) based on BIC.

```
arima(x = na.omit(data$growth_rate), order = c(3, 0, 0))

Coefficients:
         ar1     ar2      ar3  intercept
      0.3476  0.1282  -0.1122     0.0077
s.e.  0.0612  0.0644   0.0613     0.0009

sigma^2 estimated as 8.17e-05:  log likelihood = 864.45,  aic = -1718.9
AIC:  -1718.903
BIC:  -1701.042
```

```
arima(x = na.omit(data$growth_rate), order = c(1, 0, 0))

Coefficients:
         ar1  intercept
      0.3705     0.0077
s.e.  0.0572     0.0009

sigma^2 estimated as 8.344e-05:  log likelihood = 861.71,  aic = -1717.43
AIC:  -1717.43BIC:  -1706.713
```


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